

ORIGINAL RESEARCH

Biased Stackelberg game-based UAV relay anti-jamming communications: Exploiting trajectory optimization and transmission mode selection

Zhe Su¹  | Qihui Wu¹ | Nan Qi^{1,2}  | Luliang Jia³ | Zhiyong Du⁴

¹Key Laboratory of Dynamic Cognitive System of Electromagnetic Spectrum Space, Ministry of Industry and Information Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, China

²State Key Laboratory of Air Traffic Management System and Technology, and National Mobile Communications Research Laboratory, Southeast University, Nanjing, China

³School of Space Information, Space Engineering University, Beijing, China

⁴College of Information and Communication, National University of Defense Technology, Changsha, China

Correspondence

Qihui Wu, Key Laboratory of Dynamic Cognitive System of Electromagnetic Spectrum Space, Ministry of Industry and Information Technology, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, China.
Email: wuqihui2014@sina.com

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Abstract

Although unmanned aerial vehicle (UAV) relay can provide auxiliary communication due to its flexible mobility, it is vulnerable to jamming attacks. This paper considers the UAV relay anti-jamming communication issue under the threat of a malicious jammer with beam-forming jamming capability. To prevent the relay link from deteriorating, UAV trajectory adjustment and transmission mode switching between half-duplex and full-duplex are two available schemes, while they will incur the additional flying costs and continuous mode switching, respectively. To balance the trade-off between trajectory optimization and mode selection, this paper investigates the joint trajectory optimization and mode selection anti-jamming approach. First, an anti-jamming utility considering the cost-efficient and end-to-end capacity gains is designed. Second, to model the bounded rationality of both the UAV relay and the jammer due to the adversarial context, a biased Stackelberg game to analyse the competitive system interactions is proposed. Moreover, the existence of Stackelberg equilibrium (SE) in the problem is proved. Finally, a joint mode selection and trajectory optimization (JMSTO) algorithm based on the multi-armed bandit is proposed to obtain the SE. It is further demonstrated that the JMSTO algorithm has a logarithmic regret. The results show that our proposed JMSTO algorithm is superior to non-joint optimization methods.

1 | INTRODUCTION

1.1 | Background and motivation

Unmanned aerial vehicle (UAV) is a significant factor in wireless communication systems due to its flexible mobility, rapid deployment, and relatively better line-of-sight (LoS) channel between the UAV and the ground node [1–3]. UAV-assisted relaying communication is a promising technology that is widely

used for various applications, such as search and rescue [4], emergency communication [5], and computation offloading [6]. In recent years, the scenarios of UAV-assisted relaying communication works have been divided into several types based on the various requirements [7], that is, single UAV relaying networks [8], multi-user UAV-assisted relaying networks [9], multi-hop UAV relaying networks, and internet of UAVs [10]. Some studies concentrate on improving the reliability of UAV-assisted relaying communications by optimizing number of UAVs [11], UAV

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trajectory, and power [12–14], and UAV altitude [15–17]. With the development of jamming attacks, however, UAV-assisted relay communication is more susceptible to malicious jammers that sends jamming signals to disrupt UAV relay communications. The UAV may exhaust additional energy to re-transmit due to jamming attacks, which affects the performance of the UAV with the limited battery capacity [18]. Therefore, the UAV relay anti-jamming problem is still an open issue to be solved urgently.

On the one hand, the trajectory optimization anti-jamming problem is how to balance the trade-off between flying costs and jamming attacks. Although the UAV relay can reduce the jamming impacts by keeping away from the jammer, the flying costs will increase simultaneously. Conversely, jamming impacts are increased. On the other hand, half-duplex (HD) and full-duplex (FD) are the classic communication modes in wireless relay networks [19]. The UAV relay can transmit or receive based on the orthogonal channel with HD mode, but the transmission capacity is reduced. The self-interference is introduced under the FD mode due to signal leakage between the UAV relay input and output. Still, self-interference can be tolerated if the FD mode can improve the transmission capacity over the HD mode. However, the existing UAV relay anti-jamming studies often ignore communication capacity improvement through mode selection [20–24].

These two coupling optimization schemes affect the transmission capacity simultaneously. They will incur additional flying and continuous mode switching, respectively. The UAV relay can improve the transmission capacity through suitable mode selection instead of keeping away from the jammer; on the contrary, an appropriate flying trajectory can avoid mode switching frequently. Thus, how to balance the trade-off between trajectory optimization and mode selection is a challenging but meaningful research topic. Moreover, in the UAV relay anti-jamming problem, the jammer often adopts omnidirectional transmission. However, the jammer will have difficulty increasing its impact on the UAV relay due to the limited transmission power. Therefore, this paper investigates the UAV relay anti-jamming problem by joint trajectory optimization and mode selection from the cost-efficient and end-to-end capacity gains.

1.2 | Related work

The UAV, acting as a relay, can enhance the quality of information transmission over long distances. In recent years, a UAV-assisted relay communication has been widely concerned by scholars and experts [25, 26]. In ref. [25], the authors proposed a novel UAV-enabled scheduling protocol for empowering the efficiency of data dissemination in Vehicle-to-Everything networks. In ref. [26], the authors investigated a joint optimization problem of the TDMA-based user scheduling, the dynamic UAV trajectory, and the UAV transmit power to maximize the system throughput. An iterative algorithm was proposed to obtain an approximate optimal solution. However, due to the openness of its communication links, UAV-assisted

relay communication is more vulnerable to attacks from malicious jammers. To overcome possible jamming, the authors used a UAV relay to improve the communication performance of vehicular ad hoc networks (VANETs) [20]. Considering the policy hill climbing (PHC) algorithm can achieve the optimal policy without the model knowledge. A hotbooting PHC-based UAV relay strategy was proposed to help the VANET resist jamming. The authors in ref. [21] proposed a UAV relay scheme to choose the relay power against jamming attacks based on the deep Q-network and transfer learning technique. The authors in ref. [22] proposed a UAV relay scheme to improve the policy of relay power and strategies without knowing the UAV network and channel model.

The above power domain anti-jamming works neglect the mobility of the UAV relay. In the presence of a jammer, trajectory optimization in UAV relay anti-jamming is necessary. A UAV relay can balance the trade-off between the flying costs and communication performance by adjusting its position. So far, anti-jamming works for relay UAV trajectory optimization are still limited. A joint trajectory and transmit power optimization algorithm was proposed to maximize the end-to-end throughput [23]. Considering the wireless relay network with jamming signal, authors in ref. [24] optimized the flight path of the UAV relay to maximize the average signal-to-interference-plus-noise ratio (SINR) at the receiver. On the other hand, related works have investigated the problem of position-fixed mode selection. In ref. [27], the impact of jamming on the three-node two-hop cooperative amplify-and-forward communication with both HD and FD relaying was investigated. In the presence of eavesdroppers, the authors maximized the system security by using hybrid HD/FD mode [28–30]. To date, the UAV relay anti-jamming work based on joint trajectory optimization and mode selection has not been reported yet.

Game theory is an effective tool to model the dynamic interaction relationships [31]. In refs. [32] and [33], the anti-jamming problem with discrete power strategy and continuous power strategy was investigated, respectively. A Stackelberg game was formulated to model the hierarchical interactions between the user and jammer. In ref. [34], the authors investigated the multi-user power control problem in relay-assisted anti-jamming systems. A Bayesian three-layer Stackelberg game was formulated. A novel anti-intelligent UAV jamming strategy was proposed in ref. [35]. The elude problem was formulated as a Stackelberg dynamic game. The leader sub-game and followers sub-game were modelled as partially observable Markov decision process (POMDP) and Markov decision process (MDP). UAV-assisted ship anti-jamming maritime communication game was formulated in ref. [36]. In this game, the authors derived the Stackelberg equilibrium (SE) based on the channel conditions and the transmission cost coefficients.

Artificial intelligence technology has developed rapidly and become a popular research field in wireless communication [37]. One of the powerful tools is reinforcement learning, which can learn the action of malicious jamming, and then acquire the optimal anti-jamming decisions [38], multi-armed bandit (MAB) is a typical reinforcement learning method. Players choose one arm (the arm can be seen as an

optimization strategy, such as channel access strategy [39–41], power strategy [42]) to produce a reward. However, the player cannot choose the optimal arm directly without any prior knowledge. Sequential attempts to select a sequence of arms are necessary to minimize the cumulative losses [43]. Hence, the MAB model seeks the balance between ‘exploration’ and ‘exploitation’.

More recent work has considered the MAB model in ground node anti-jamming but few in UAV relay anti-jamming. In ref. [39], the authors studied the anti-jamming vehicle-to-vehicle communication. An MAB problem was called upon to perform the channel-selection process. In ref. [40], the anti-jamming multi-channel access in cognitive radio networks was formulated as a non-stochastic multiple-armed bandit problem. The secondary sender and receiver chose their channels to maximize the throughput. In ref. [41], the authors investigated the multi-domain anti-jamming problem, in which the Stackelberg game and MAB were formulated in the power domain and spectrum domain, respectively. In ref. [42], the discrete power control problem in anti-jamming relay communication networks was formulated as an MAB problem. Further, based on the Upper Confidence Bound (UCB1) method [44–46], the discrete power control online learning algorithm was proposed to update the power strategy. The existing Stackelberg game models assume that the leader and follower are perfectly rational and play the game optimally. Unfortunately, in the adversarial context, the perfect rationality of the UAV relay and jammer is unreasonable. The leader and follower may have characteristics of human decision-makers, and their decisions will deviate from the expected optimal strategies because of bounded rationality. The proposed Stackelberg game combines the advantages of the MAB method, prompting us to develop this work.

1.3 | Contributions

To the best of the authors’ knowledge, this paper is the first to investigate the UAV relay joint trajectory optimization and mode selection anti-jamming communication issue under the threat of a malicious jammer with beam-forming jamming capability. An anti-jamming utility considering the cost-efficient and end-to-end capacity gains is designed. The bounded rationality of both the UAV relay and the jammer is modelled because the UAV relay and jammer may have characteristics of human decision-makers in an adversarial context. A biased Stackelberg game is proposed to analyse competitive interactions. In this game, the UAV relay acts as a leader and takes action first, while the jammer acts as a follower. The joint mode selection and trajectory optimization (JMSTO) algorithm based on the MAB is proposed to obtain the SE. Simulations are made to verify the effectiveness of the proposed algorithm. The specific contributions of our work are summarized as follows.

- This paper considers the UAV relay anti-jamming communication issue under the threat of a malicious jammer with beam-forming jamming capability. In order to balance the trade-off between trajectory optimization and

TABLE 1 Notations

Notations	Description
p_s, p_r, p_j	Transmission power at source node, UAV relay, and jammer
$g_{i,r,k}$	Channel gain of node i to UAV relay in stage k
$g_{r,k}$	Channel gain of loop channel gain in stage k
N_0	Background noise
$(x_s, y_s, 0)$	Coordinates of source node
$(x_d, y_d, 0)$	Coordinates of destination node
$(x_j, y_j, 0)$	Coordinates of jammer
α	Path-loss exponent coefficient
η	Flying costs per unit distance
ε	NLoS additional attenuation factor

mode selection, we propose the JMSTO anti-jamming approach.

- An anti-jamming utility considering the cost-efficient and end-to-end capacity gains is designed. To model the bounded rationality of both the UAV relay and the jammer due to the adversarial context, we propose a biased Stackelberg game to analyse the competitive system interactions. And then, we prove the existence of SE and propose the JMSTO algorithm based on the MAB to obtain SE. In addition, the regret of the proposed algorithm is proved to be logarithmic.
- Simulation results show the impacts of bounded rationality, flying costs, and beam angle width on average utility. By comparing the performance of the proposed JMSTO algorithm, fixed trajectory (FT), fixed transmission mode (FTM), and random trajectory and random mode (RTRM) methods, the advantage of our algorithm is demonstrated.

The rest of this paper is organized as follows. In Section 2, the UAV relay anti-jamming model, and the problem formulation are given. The biased Stackelberg game, the existence of SE, and the JMSTO algorithm are presented in Section 3. Simulation results show the impacts of bounded rationality, flying costs, and beam angle width on average utility in Section 4. Besides, average utilities under the JMSTO algorithm, FT, FTM, and RTRM methods are compared. Finally, Section 5 concludes this work.

Notations: related notations are listed in Table 1.

2 | SYSTEM MODEL AND PROBLEM FORMULATION

2.1 | System model

As shown in Figure 1, we consider the UAV relay anti-jamming scenario, where the UAV relay flies from the starting point to the end point. A UAV acts as an aerial relay (r) and assists data transmission from the source node (s) to the destination node (d) at a fixed altitude h_r . To simplify the trajectory optimization problem, we define the flying stage set as $\mathcal{K} = [1, 2, \dots, K]$,

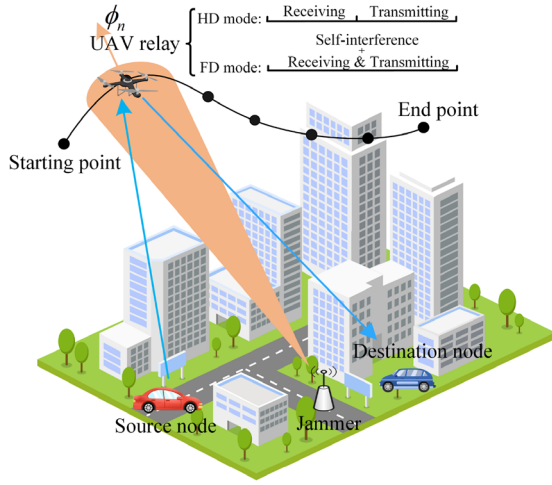


FIGURE 1 UAV relay anti-jamming scenario

where K represents the total number of flying stages. In each stage, the set of discretized positions is described as $\mathcal{A}_{r_1,k} = [tr_{1,k}, tr_{2,k}, \dots, tr_{L,k}]$, $k \in \mathcal{K}$, where L represents the total number of discretized positions. Therefore, the trajectory of UAV relay can be seen as a combination of discrete positions. Moreover, the decode-and-forward (DF) protocol is adopted in this paper. UAV relay needs to operate HD or FD transmission modes adaptively in each stage to improve the end-to-end transmission capacity. The UAV relay joint strategy set is given as

$$\mathcal{A}_{r,k} = \mathcal{A}_{r_1,k} \times \mathcal{A}_{r_2}, \quad (1)$$

where $\mathcal{A}_{r_2} = [HD, FD]$ represents the communication mode set.

Based on the antenna-array beam-forming technology, the jammer (j) has targeting ability and adopts the beam-forming transmission to jam the UAV relay. We assume that the jammer only attacks the UAV relay transmission in this work. The reason is that the destination node cannot receive the forwarded information if the jammer attacks the UAV relay effectively. In each stage, the jammer selects different strategy from jamming beam direction set $\mathcal{A}_j = [\phi_1, \phi_2, \dots, \phi_N]$, where N represents the number of available azimuths.

We assume that the LoS probability of the air-to-ground (A2G) channel is calculated as [47]

$$P(LoS)_{i,r,k} = \frac{1}{1 + A \exp(-B[\xi_{i,r,k} - A])}, \quad (2)$$

where the values of A and B rely on the scenario environment [47, 48], such as suburban, urban, and dense urban. Then, the elevation angle $\xi_{i,r,k}$ between UAV relay and node i ($i = s, d, j$) is defined as

$$\xi_{i,r,k} = \frac{180}{\pi} \arcsin\left(\frac{h_r}{d_{i,r,k}}\right). \quad (3)$$

The distance from node i to the UAV relay is given as

$$d_{i,r,k} = \sqrt{(x_{r,k} - x_i)^2 + (y_{r,k} - y_i)^2 + h_r^2}. \quad (4)$$

The non-line-of-sight (NLoS) probability of the A2G channel can be obtained as

$$P(NLoS)_{i,r,k} = 1 - P(LoS)_{i,r,k}. \quad (5)$$

Inspired by refs. [49, 50], we introduce the NLoS additional attenuation factor ε . The expression for the channel power gain from the UAV relay to the ground node is

$$g_{i,r,k} = P(LoS)_{i,r,k} \times d_{i,r,k}^{-\alpha} + P(NLoS)_{i,r,k} \times \varepsilon d_{i,r,k}^{-\alpha}. \quad (6)$$

2.2 | Problem formulation

The UAV relay anti-jamming problem is divided into two steps. In step 1, the source node transmits signals to the UAV relay, while the jammer senses the broadcast signals and tries to block the UAV relay. In step 2, the UAV relay applies the DF protocol and forwards the decoding signals to the destination node. The optimization problems under the HD mode and the FD mode are discussed below.

2.2.1 | UAV relay communication under the HD mode

UAV relay with HD mode forwards signals to the destination node without being affected by the self-interference. Thus, in the first hop, SINR received at UAV relay is defined as

$$\Gamma_{s,r,k}^{HD} = \frac{g_{s,r,k} p_s}{N_0 + g_{j,r,k} p_j \delta(\phi_{n,k}, l_k)}. \quad (7)$$

An indicator function $\delta(\phi_{n,k}, l_k)$ is denoted as

$$\delta(\phi_{n,k}, l_k) = \begin{cases} 1, & l_k \text{ is located in } \phi_{n,k}, \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

where $l_k \in \mathcal{A}_{r_1,k}$ and $\phi_{n,k} \in \mathcal{A}_j$. $\delta(\phi_{n,k}, l_k)$ takes value 1 if the UAV relay is located in the jamming beam area, otherwise it is 0.

In the second hop, the signal-to-noise ratio (SNR) receiver at the destination node can be given by

$$\Gamma_{d,r,k}^{HD} = \frac{g_{d,r,k} p_r}{N_0}. \quad (9)$$

Inspired by ref. [51], based on the cost-efficient perspective, we introduce the flying costs to constrain the redundant flying distance of the UAV relay, which is formulate as

$$E_k = \eta \sqrt{(x_{r,k} - x_{r,k-1})^2 + (y_{r,k} - y_{r,k-1})^2}. \quad (10)$$

UAV relay's utility function in HD mode is expressed as

$$C_{r,k}^{\text{HD}} = \frac{1}{2} \log_2 \left(1 + \min \left\{ \Gamma_{s,r,k}^{\text{HD}}, \Gamma_{d,r,k}^{\text{HD}} \right\} \right) - E_k. \quad (11)$$

For simplified expression, (11) is rewritten as

$$C_{r,k}^{\text{HD}} = \frac{1}{2} \omega^{\text{HD}} - E_k, \quad (12)$$

where $\omega^{\text{HD}} \triangleq \log_2 (1 + \min \{ \Gamma_{s,r,k}^{\text{HD}}, \Gamma_{d,r,k}^{\text{HD}} \})$.

2.2.2 | UAV relay communication under the FD mode

In the FD transmission mode, the UAV relay only needs one transmission channel, which introduces the self-interference. Thus, for the first hop, the SINR received at the UAV relay is defined as

$$\Gamma_{s,r,k}^{\text{FD}} = \frac{g_{s,r,k} p_s}{N_0 + g_{r,k} p_r + g_{j,r,k} p_j \delta(\phi_{n,k}, l_k)}. \quad (13)$$

In the second hop, the SNR received at the destination node can be given by

$$\Gamma_{d,r,k}^{\text{FD}} = \frac{g_{d,r,k} p_r}{N_0}. \quad (14)$$

UAV relay's utility function in FD mode is expressed as

$$C_{r,k}^{\text{FD}} = \log_2 \left(1 + \min \left\{ \Gamma_{s,r,k}^{\text{FD}}, \Gamma_{d,r,k}^{\text{FD}} \right\} \right) - E_k. \quad (15)$$

Similarly, (15) is rewritten as

$$C_{r,k}^{\text{FD}} = \omega^{\text{FD}} - E_k, \quad (16)$$

where $\omega^{\text{FD}} \triangleq \log_2 (1 + \min \{ \Gamma_{s,r,k}^{\text{FD}}, \Gamma_{d,r,k}^{\text{FD}} \})$.

If the UAV relay's utility value in FD mode is greater than its utility value in HD mode, it will forward the message using HD mode. Otherwise, the UAV relay selects the FD mode. The utility function of the UAV relay* is described as

$$U_{r,k} = \begin{cases} C_{r,k}^{\text{HD}}, & \text{if } C_{r,k}^{\text{HD}} > C_{r,k}^{\text{FD}}, \\ C_{r,k}^{\text{FD}}, & \text{otherwise.} \end{cases} \quad (17)$$

From the perspective of the jammer, the purpose of the jammer is to degrade the UAV relay's communication performance. Thus, the utility function of the jammer is denoted as

$$U_{j,k} = \begin{cases} -\omega^{\text{HD}}, & \text{UAV relay selects HD mode,} \\ -\omega^{\text{FD}}, & \text{UAV relay selects FD mode.} \end{cases} \quad (18)$$

* The discussion for performance evaluation using additional metric, such as average transmission rates, can be found in APPENDIX.

3 | THE BIASED STACKELBERG GAME SOLUTION AND JMSTO ALGORITHM

Here, the interactions between the UAV relay and the jammer are formulated as a biased Stackelberg game. Then, the existence of SE is proved. Finally, the JMSTO algorithm is proposed to obtain the SE.

3.1 | Biased Stackelberg game

In this paper, bounded rationality is introduced to describe the biased decisions of the UAV relay and jammer [52, 53]. Furthermore, the global biased Stackelberg game is designed to model the hierarchical relationships between the UAV relay and jammer. We assume that the UAV relay acts as a leader and the jammer acts as a follower. The global biased Stackelberg game is formulated as $\mathcal{G} = \{r, j, \mathcal{A}_r, \mathcal{A}_j, U_r, U_j\}$, where r and j represent the UAV relay and jammer, respectively. $\mathcal{A}_r$ and $\mathcal{A}_j$ represent the UAV relay's and jammer's strategy set, respectively. U_r and U_j represent the UAV relay's and jammer's utility, respectively.

The rationality factor is introduced for each stage to describe the rationality degree. A higher value of the rationality factor means a higher rationality degree. The rationality factor $\beta_j \in [0, 1]$ is introduced due to the follower's bounded rationality. The utility function of the UAV relay is rewritten as

$$\hat{U}_{r,k} = \sum_{m=1}^{2L} \lambda_m U_{r,k}, \quad (19)$$

where λ_m represents the probability that the UAV relay strategy $\phi_{m,k}$ ($\phi_{m,k} \in \mathcal{A}_{r,k}$) is regarded as $\phi'_{m,k}$ ($\phi'_{m,k} \in \mathcal{A}_{r,k} \setminus \phi_{m,k}$), and is defined as

$$\lambda_m = \frac{1 - \beta_j}{2L - 1}. \quad (20)$$

Then, the rationality factor $\beta_r \in [0, 1]$ is introduced due to the leader's bounded rationality. The utility function of the jammer is rewritten as

$$\hat{U}_{j,k} = \sum_{n=1}^N \gamma_n U_{j,k}, \quad (21)$$

where γ_n represents the probability that the jammer strategy $\phi_{n,k}$ ($\phi_{n,k} \in \mathcal{A}_j$) is regarded as $\phi'_{n,k}$ ($\phi'_{n,k} \in \mathcal{A}_j \setminus \phi_{n,k}$), and is defined as

$$\gamma_n = \frac{1 - \beta_r}{N - 1}. \quad (22)$$

The objective function of the UAV relay and jammer are expressed as

$$\max_{\varphi_{m,k}} \hat{U}_{r,k}(\varphi_{m,k}, \phi_{n,k}), \quad (23)$$

$$\max_{\phi_{n,k}} \hat{U}_{j,k}(\varphi_{m,k}, \phi_{n,k}). \quad (24)$$

The joint trajectory optimization and mode selection anti-jamming problem can be solved using the proposed biased Stackelberg game, which is illustrated as

$$\left\{ \begin{array}{l} \max_{\varphi_{m,k}} \hat{U}_{r,k}(\varphi_{m,k}, \phi_{n,k}) \\ \text{subject to : } \varphi_{m,k} \in \mathcal{A}_{r,k} \\ \text{The optimal solution : } (\varphi_{n,k}^*) \\ \max_{\phi_{n,k}} \hat{U}_{j,k}(\varphi_{m,k}, \phi_{n,k}) \\ \text{subject to : } \phi_{n,k} \in \mathcal{A}_j \\ \text{The optimal solution : } (\varphi_{m,k}^*). \end{array} \right. \quad (25)$$

3.2 | The existence of SE

Motivated by refs. [32] and [54], the SE can be defined as follows.

Definition 1. If neither the UAV relay nor the jammer can improve the utility by deviating its optimal strategy unilaterally, the strategy pair $(\varphi_{m,k}^*, \phi_{n,k}^*)$ constitutes the SE when the following conditions hold.

$$\hat{U}_{r,k}(\varphi_{m,k}^*, \phi_{n,k}^*) \geq \hat{U}_{r,k}(\varphi_{m,k}, \phi_{n,k}^*), \quad (26)$$

$$\hat{U}_{j,k}(\varphi_{m,k}^*, \phi_{n,k}^*) \geq \hat{U}_{j,k}(\varphi_{m,k}^*, \phi_{n,k}), \quad (27)$$

where $\varphi_{m,k}^*$ and $\phi_{n,k}^*$ represent optimal strategies of the UAV relay and the jammer, respectively.

Lemma 1. *There exists a UAV relay's stationary strategy and a smart jammer's stationary strategy, which constitute an SE.*

Proof1. According to ref. [54], the jammer aims to maximize the utility and makes strategy based on the best response

$$\phi_{n,k}^* = \arg \max \hat{U}_{j,k}(\varphi_{m,k}, \phi_{n,k}). \quad (28)$$

The UAV relay can obtain the optimal strategy after sensing the jammer's optimal strategy

$$\varphi_{m,k}^* = \arg \max \hat{U}_{r,k}(\varphi_{m,k}, \phi_{n,k}^*(\varphi_{m,k})). \quad (29)$$

Based on this, strategy pair $(\varphi_{m,k}^*, \phi_{n,k}^*)$ constitutes the SE.

3.3 | The JMSTO algorithm

We propose the JMSTO algorithm based on the MAB method to achieve the SE. In the JMSTO algorithm, the UAV relay and jammer can choose their strategies based on the historical action and reward. Due to the hierarchical relationships between the UAV relay and jammer, the JMSTO algorithm consists of the leader JMSTO algorithm and the follower JMSTO algorithm.

3.3.1 | Leader JMSTO algorithm

For the UAV relay, we define the times that $\varphi_{m,k}$ has been selected in the past T epochs as

$$num_{r,k}(T) = \sum_{t=1}^T f_{r,k}(\varphi_{m,k}, \varphi'_{m,k}(t)), \quad (30)$$

where $\varphi'_{m,k}(t)$ denotes which arm is selected by the UAV relay in the t th epoch. if $\varphi_{m,k} = \varphi'_{m,k}(t)$ holds, $f_{r,k}(\varphi_{m,k}, \varphi'_{m,k}(t)) = 1$; otherwise, $f_{r,k}(\varphi_{m,k}, \varphi'_{m,k}(t)) = 0$.

The statistical average reward when the UAV relay selects the $\varphi_{m,k}$ in the past T epochs is defined as

$$\mu_{r,k} = \frac{\sum_{t=1}^T \hat{U}_{r,k}(t) f_{r,k}(\varphi_{m,k}, \varphi'_{m,k}(t))}{num_{r,k}(T)}. \quad (31)$$

The total utility of UAV relay in the past T epochs is defined as

$$\hat{U}'_{r,k}(T) = \sum_{t=1}^T \hat{U}_{r,k}(t). \quad (32)$$

Regret is an essential metric for evaluating the performance of the MAB algorithm, which describes the reward loss due to the player not selecting the optimal arm [55]. We can express the regret of the UAV relay as

$$E[R_{r,\varphi_{m,k}}(T)] = E[\hat{U}'_{r,\varphi_{m,k}^*}(T)] - E[\hat{U}'_{r,\varphi_{m,k}}(T)], \quad (33)$$

where $\hat{U}'_{r,\varphi_{m,k}^*}(T)$ denotes UAV relay's accumulative utility when it selects optimal arm in the past T epochs.

3.3.2 | Follower JMSTO algorithm

For the jammer, we define the times that $\phi_{n,k}$ has been selected in the past T epochs is

$$num_{j,k}(T) = \sum_{t=1}^T f_{j,k}(\phi_{n,k}, \phi'_{n,k}(t)), \quad (34)$$

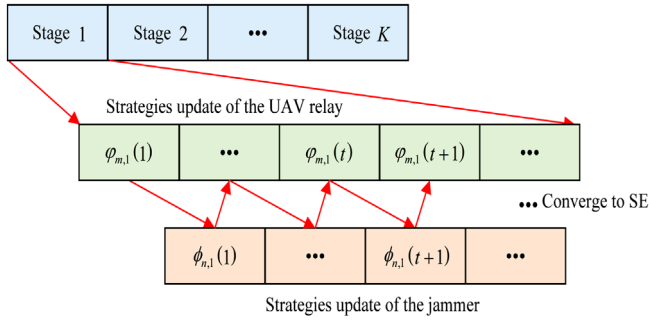


FIGURE 2 The implementation process of the SE

where $\phi'_{n,k}(t)$ denotes which arm is selected by the jammer in the t th epoch. If $\phi_{n,k} = \phi'_{n,k}(t)$ holds, $f_{j,k}(\phi_{n,k}, \phi'_{n,k}(t)) = 1$; otherwise, $f_{j,k}(\phi_{n,k}, \phi'_{n,k}(t)) = 0$.

The statistical average reward when the jammer selects the $\phi_{n,k}$ in the past T epochs is defined as

$$\mu_{j,k} = \frac{\sum_{t=1}^T \hat{U}_{j,k}(t) f_{j,k}(\phi_{n,k}, \phi'_{n,k}(t))}{num_{j,k}(T)}. \quad (35)$$

The total utility of the jammer in the past T epochs is defined as

$$\hat{U}'_{j,k}(T) = \sum_{t=1}^T \hat{U}_{j,k}(t). \quad (36)$$

Further, the regret of the jammer can be expressed as

$$E[R_{j,\phi_{n,k}}(T)] = E\left[\hat{U}'_{j,\phi_{n,k}^*}(T)\right] - E\left[\hat{U}'_{j,\phi_{n,k}}(T)\right], \quad (37)$$

where $\hat{U}'_{j,\phi_{n,k}^*}(T)$ denotes jammer's accumulative utility if it selects optimal arm in the past T epochs.

In this work, minimizing the regret values are equivalent to maximizing the long-term reward. Based on the UCB1 method [45], UAV relay joint strategy and jammer strategy can be updated according to (38) and (39), respectively.

$$\varphi_{m,k}(T+1) = \arg \max_{m \in [1,2,\dots,2L]} \left(\mu_{r,k} + \sqrt{\frac{2 \ln T}{num_{r,k}(T)}} \right), \quad (38)$$

$$\phi_{n,k}(T+1) = \arg \max_{n \in [1,2,\dots,N]} \left(\mu_{j,k} + \sqrt{\frac{2 \ln T}{num_{j,k}(T)}} \right). \quad (39)$$

The JMSTO algorithm is summarized in Algorithm 1.

The process of SE is depicted in Figure 2. For each stage, the UAV relay and jammer are optimized in different layers and iterate with each other. After several epochs, both the UAV relay and jammer can obtain the equilibrium. It indicates that the proposed JMSTO algorithm can converge to the SE, which can maximize the network utility.

ALGORITHM 1 JMSTO algorithm

Input: Source node power p_s , UAV relay power p_r , jammer power p_j , flying costs parameter η , background noise N_0 , total number of epochs $t_{\max}$, total number of stages K , strategy sets $\mathcal{A}_r$ and $\mathcal{A}_j$.

Output: $\{\varphi_{m,1}^*, \varphi_{m,2}^*, \dots, \varphi_{m,K}^*\}, \{\phi_{n,1}^*, \phi_{n,2}^*, \dots, \phi_{n,K}^*\}$.

```

1: for  $k = 0$  to  $K$  do
2:    $k = k + 1$ .
3:   for  $t = 0$  to  $t_{\max}$  do
4:      $t = t + 1$ .
5:     UAV relay selects the optimal arm according to (38).
6:     jammer selects the optimal arm according to (39).
7:     obtain UAV relay's utility (17) and jammer's utility (18).
8:     update  $num_{r,k}(t)$ ,  $num_{j,k}(t)$ ,  $\mu_{r,k}$ , and  $\mu_{j,k}$ , respectively.
9:     record the UAV relay's and jammer's selected arm.
10:    return to 4 until  $t \geq t_{\max}$ .
11:   end for
12:   return to 2 until  $k \geq K$ .
13: end for

```

3.4 | The convergence analysis of JMSTO algorithm

Minimizing the regret values are equivalent to maximizing the long-term reward. It is possible to demonstrate the convergence of the JMSTO algorithm by showing that it has logarithmic regret. The details can be found as follows.

Lemma 2. The expected regret of the UCB1-based JMSTO algorithm has a logarithmic regret, that is, $E[R(T)] \sim O(\log T)$. Its regret is upper bounded as

$$E[R(T)] \leq \sum_{q: \hat{U}'_q < \hat{U}'_{q^*}} \Delta_q \left(\frac{8 \ln T}{\Delta_q^2} + 1 + \frac{\pi^2}{3} \right). \quad (40)$$

Proof 2. Motivated by ref. [45, Proof of Theorem 1], we have

$$E[num(T)] \leq \frac{8 \ln S(T)}{\Delta_q^2} + 1 + \frac{\pi^2}{3}, \quad (41)$$

where $\Delta_q = \hat{U}'_{q^*} - \hat{U}'_q$. $S(T)$ is the number of strategy switches that occurred throughout T epochs.

Similar to refs. [41, 56], we can obtain the upper bound of the expected regret by using the given equations

$$\begin{aligned}
E[R(T)] &\leq \sum_{q: \hat{U}'_q < \hat{U}'_{q^*}} (\hat{U}'_{q^*} - \hat{U}'_q) E[num(T)] \\
&\leq \sum_{q: \hat{U}'_q < \hat{U}'_{q^*}} \Delta_q \left(\frac{8 \ln S(T)}{\Delta_q^2} + 1 + \frac{\pi^2}{3} \right)
\end{aligned}$$

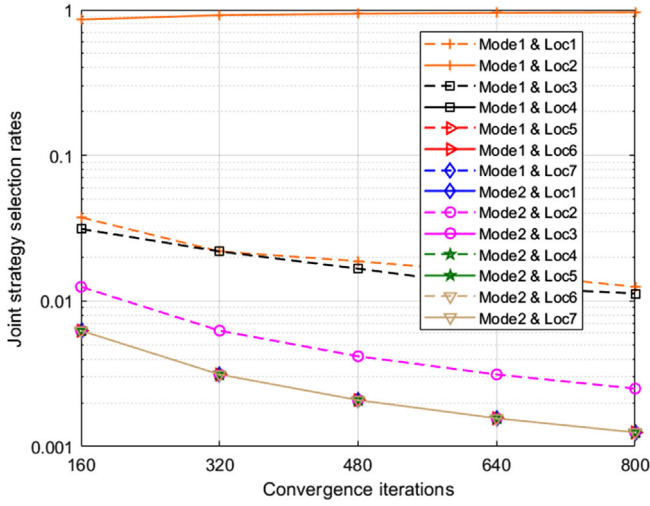


FIGURE 3 The convergence iterations of UAV relay

$$\leq \sum_{q: \hat{D}_q' < \hat{D}_{q^*}'} \Delta_q \left(\frac{8 \ln T}{\Delta_q^2} + 1 + \frac{\pi^2}{3} \right). \quad (42)$$

4 | NUMERICAL RESULTS

This section gives the simulation parameters settings and the position information (including source node, destination node, jammer, UAV relay's starting point, and UAV relay's end point). Simulation results and brief analyses are further presented.

Specifically, parameters are set as: $\alpha = 2$, $p_s = 2W$, $p_r = 3W$, $p_j = 6W$, $N_0 = 10^{-6}W$, $A = 9.6117$ [57], $B = 0.1139$ [57], $\varepsilon = 20\text{dB}$ [49], $K = 5$, $t_{\max} = 800$, and $\eta = 0.04$. The set of jamming beam direction is defined as: $\mathcal{A}_j = [0^\circ - 40^\circ, 40^\circ - 80^\circ, \dots, 320^\circ - 360^\circ]$.

4.1 | Convergence analysis for UAV relay and jammer

Since the convergence results in each flying stage are similar, we take stage 1 as an example. Figures 3 and 4 show the strategy selection rates of the UAV relay and the jammer, respectively. The selection rates describe the ratio of the selection number for each strategy to the total selection number.

As shown in Figure 3, 'Mode1' and 'Mode2' denote HD mode and FD mode, respectively. 'Loc1', 'Loc2', 'Loc3', 'Loc4', 'Loc5', 'Loc6', and 'Loc7' denote available positions in each stage. With the increasing iteration numbers, the UAV relay tends to select joint strategy 'Mode1 + Loc2', that is, UAV relay flies to the coordinate $(150m, 300m, 200m)$, and forwards the signals by switching the HD mode. Specifically, the joint strategy selection rates of UAV relay can be found in Table 2, where the bolded values are the highest selection rates for each number of iterations.

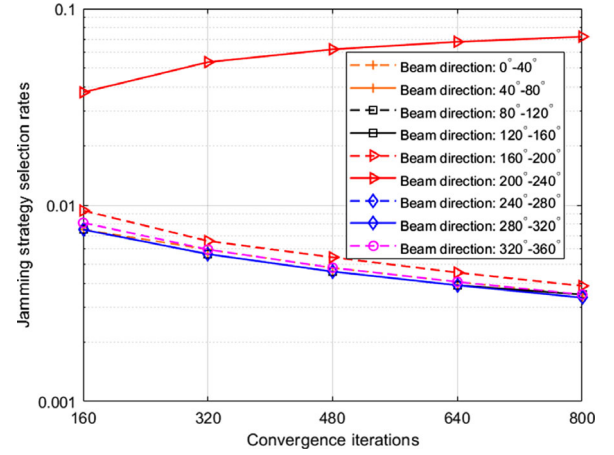


FIGURE 4 The convergence iterations of jammer

In Figure 4, similarly, jammer tends to select the beam direction during communication process, that is, $200^\circ - 240^\circ$. Note that both UAV relay and jammer may select other strategies with some selection rates, and this behaviour can be seen as an 'exploration' of potential strategies. Specifically, the jamming strategy selection rates of jammer can be found in Table 3, where the bolded values are the highest selection rates for each number of iterations.

4.2 | Flying trajectory of UAV relay

The coordinates of different nodes are presented in three-dimension space. We assume that the UAV relay and jammer execute the joint strategy and jamming strategy with the highest selection rate in each stage. The details are shown in Figure 5. The coordinates of source node, destination node, and jammer are $(150m, -200m, 0m)$, $(900m, 300m, 0m)$, and $(450m, 0m, 0m)$, respectively. UAV relay flies from the starting point $(0m, 0m, 200m)$ to the end point $(900m, -200m, 200m)$. The flying process is divided into $K = 5$ stages. For each stage, the UAV relay has $S = 14$ joint position and transmission mode strategies to select. The adjacent distance of each stage is $150m$.

4.3 | Regret values of UAV Relay and jammer with JMSTO algorithm

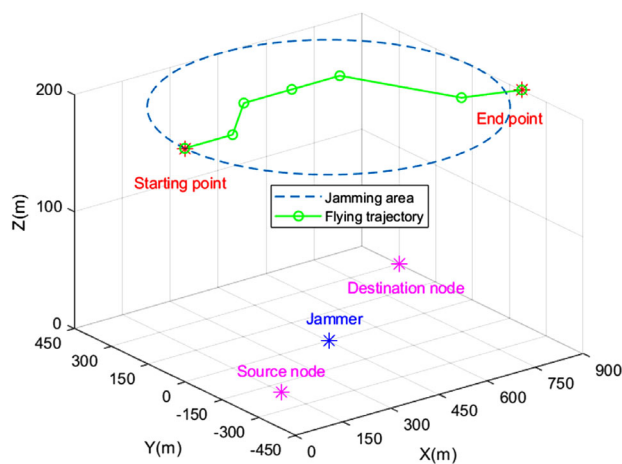
Figure 6 shows the regret values of the UAV relay and jammer with the JMSTO algorithm. Taking stage 1 as an example, we can find that the regret values increase rapidly at the beginning of the iterations. After 160 iterations, the regret values increase slowly. The results show that the jammer and UAV relay adopt the JMSTO learning algorithm to select their strategies. The reward loss decreases with the convergence iterations because the number of non-optimal strategy choices decreases.

TABLE 2 Joint strategy selection rates

	160th iteration		320th iteration		480th iteration		640th iteration		800th iteration	
	Mode 1	Mode 2	Mode 1	Mode 2	Mode 1	Mode 2	Mode 1	Mode 2	Mode 1	Mode 2
Loc1	3.75%	0.63%	2.19%	0.31%	1.88%	0.21%	1.56%	0.16%	1.25%	0.13%
Loc2	85.63%	1.25%	91.88%	0.63%	93.96%	0.42%	95.31%	0.31%	96.13%	0.25%
Loc3	3.13%	0.63%	2.19%	0.31%	1.67%	1.56%	1.25%	0.16%	1.13%	0.13%
Loc4	0.63%	0.63%	0.31%	0.31%	0.21%	0.21%	0.16%	0.16%	0.13%	0.13%
Loc5	0.63%	0.63%	0.31%	0.31%	0.21%	0.21%	0.16%	0.16%	0.13%	0.13%
Loc6	0.63%	0.63%	0.31%	0.31%	0.21%	0.21%	0.16%	0.16%	0.13%	0.13%
Loc7	0.63%	0.63%	0.31%	0.31%	0.21%	0.21%	0.16%	0.16%	0.13%	0.13%

TABLE 3 Jamming strategy selection rates

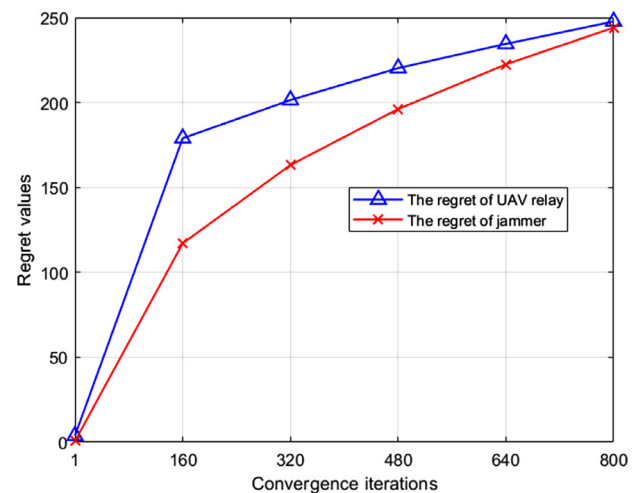
	160th iteration	320th iteration	480th iteration	640th iteration	800th iteration
Beam direction 0° – 40°	7.50%	5.94%	4.79%	4.06%	3.50%
Beam direction 40° – 80°	7.50%	5.63%	4.58%	3.91%	3.38%
Beam direction 80° – 120°	7.50%	5.63%	4.58%	3.91%	3.50%
Beam direction 120° – 160°	7.50%	5.63%	4.58%	3.91%	3.50%
Beam direction 160° – 200°	9.38%	6.56%	5.42%	4.53%	3.88%
Beam direction 200° – 240°	37.50%	53.44%	62.08%	67.81%	72.00%
Beam direction 240° – 280°	7.50%	5.63%	4.58%	3.91%	3.38%
Beam direction 280° – 320°	7.50%	5.63%	4.58%	3.91%	3.38%
Beam direction 320° – 360°	8.13%	5.94%	4.79%	4.06%	3.50%

**FIGURE 5** Nodes coordinates and flying trajectory

4.4 | Average utility comparison of the UAV relay and jammer with different rationality factors

We analyse the average utility comparison of the UAV relay and jammer with different rationality factors in stage 1.

Figure 7 shows the impact of the rationality factor (β_j) on the UAV relay's average utility. Increasing rationality factor degrades

**FIGURE 6** The regret values of UAV relay and jammer

the improvement of the UAV relay's average utility. The reason is that the jammer may not choose an optimal joint strategy, which can be seen as the jammer making mistakes. Moreover, we can find that the average utility will degrade with the growth of flying costs.

Figure 8 shows the rationality factor (β_j) on the jammer's average utility. Similarly, the UAV relay deviates from its optimal strategy, and bounded rationality increases the jammer's average

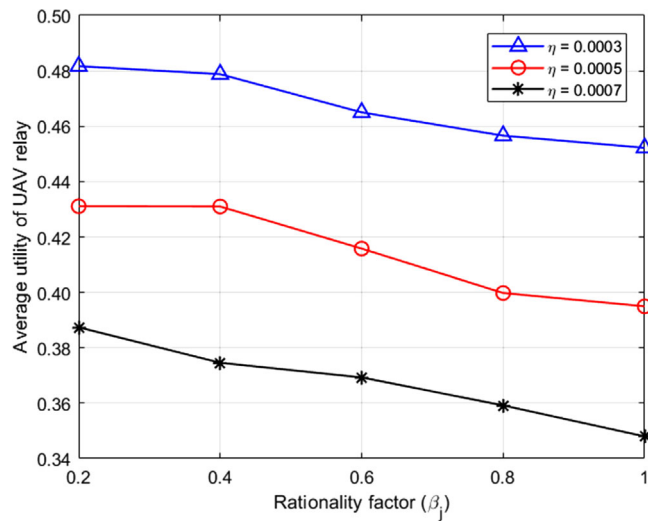


FIGURE 7 UAV relay's average utility comparison

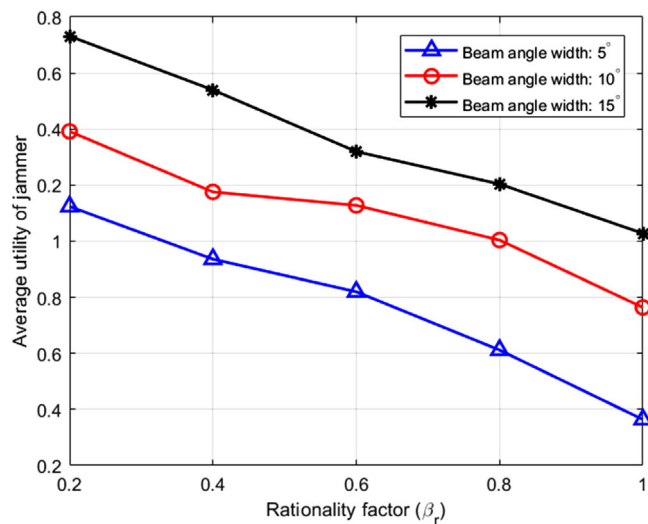


FIGURE 8 Jammer's average utility comparison

utility. In addition, the larger the beam angle width is, the greater the average utility of the jammer. The reason is that the larger beam angle width can easily cover the UAV relay position.

4.5 | UAV relay's average utility comparison with different methods

By comparing the performance of the proposed JMSTO algorithm with the FT, FTM, and RTRM methods in Figure 9. Under the FT method, flying trajectory is fixed (i.e. (150 m, 0 m), (300 m, -150 m, 200 m), (450 m, -150 m, 200 m), (600 m, -150 m, 200 m), and (750 m, -150 m, 200 m)), meanwhile the transmission mode is optimized. Under the FTM method, the UAV relay mode is fixed and the flying trajectory is optimized. Under the RTRM method, the UAV relay randomly

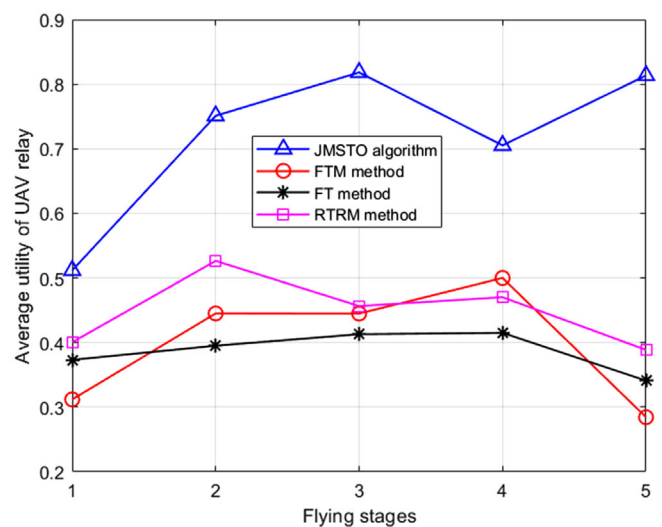


FIGURE 9 Utility comparison with different algorithms

selects the flying trajectory and transmission mode. We can see that the average utility of the proposed JMSTO algorithm is better than FT, FTM, and RTRM methods.

5 | CONCLUSION

This paper has considered the UAV relay anti-jamming communication issue under the threat of a malicious jammer with beam-forming jamming capability. UAV trajectory adjustment and transmission mode switching are two schemes. They are coupled and affect the transmission capacity simultaneously. To this end, the joint trajectory optimization and mode selection anti-jamming approach has been proposed to balance the trade-off between trajectory optimization and mode selection. First, an anti-jamming utility considering the cost-efficient and end-to-end capacity gains has been designed. Second, to model the bounded rationality of both the UAV relay and the jammer due to the adversarial context, a biased Stackelberg game has been proposed to analyse the competitive system interactions. Moreover, the SE has been proposed. The regret of the proposed algorithm is proved to be logarithmic. Finally, the JMSTO algorithm based on the MAB has been proposed to obtain the SE. The superiority of the proposed JMSTO algorithm has been verified.

AUTHOR CONTRIBUTIONS

Zhe Su: Data curation, investigation, software, writing - original draft, writing - review and editing. Qihui Wu: Supervision. Nan Qi: Supervision, writing - review and editing. Luliang Jia: Supervision, writing - review and editing. Zhiyong Du: Supervision, writing - review and editing.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Zhe Su  <https://orcid.org/0000-0002-2987-3484>

Nan Qi  <https://orcid.org/0000-0002-0125-370X>

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APPENDIX A

The performance evaluation with average transmission rates is discussed as follows.

By comparing the average transmission rates of the proposed JMSTO algorithm with the FT (the fixed flying trajectory is the same as Section 4.5), FTM, and RTRM methods in Table A1, the result demonstrates that our proposed JMSTO algorithm is also superior to non-joint optimization methods. Throughout all flying stages, the average transmission rates of UAV relay with the proposed JMSTO algorithm is greater than non-joint optimization methods 67.49% (FTM method), 79.72% (FT method), and 47.88% (RTRM method), respectively (note that we make a normalization on the channel bandwidth).

TABLE A1 Average transmission rates of UAV relay

	JMSTO algorithm	FTM method	FT method	RTRM method
Flying stage 1	0.5740	0.3545	0.3962	0.4301
Flying stage 2	0.8263	0.5085	0.4048	0.5897
Flying stage 3	0.7216	0.4800	0.4119	0.5028
Flying stage 4	0.7353	0.5475	0.4128	0.5286
Flying stage 5	0.8241	0.3075	0.4230	0.4383

1The values in bold are the average highest transmission rates for each flying stage